COIT11237

LECTURE NOTES

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Document Start Date: 11th of June 2026
Latest Update: June 11, 2026

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1 Introduction to Databases

1.1 What is MySQL?:

Structured Query Language (SQL) is the language used to manage relational database management systems (RDBMS). It follows the CRUD principles. In that it Create, Replace, Update, & and Deletes. SQL is used in both small & large scale applications.

1.2 Databases:

A database (DB) can be made up of multiple tables. The conventional naming is singular. For examples, a customers' information DB would just be named customer, not customers. The table also needs a primary key (PK) as a unique identifier. This is so a singular record can be found. Tables can also be linked together through foreign keys (FK) which is just a PK in another table. When this is done, a relational database is created.

1.3 Problems with Spreadsheets

1. Have redundant/wrong/out-of-date data present within the sheet. This can create conflicting copies throughout.
2. Multiple themes per row creates modification problems
    - Insertion anomalies
    - Deletion anomalies

1.4 Database System Components:

- Users use forms/reports to modify data
- Database application converts user inputs into SQL data, which is then sent to DBMS
- DBMS:
    - Processes the SQL
    - Enforces the DB rules
    - Provides concurrency, security, back-up and recovery
- Database is where the data is physically stored

1.5 SQL:

SQL is an international standard for the talking with DBs and their tables. For example:

```
SELECT *
FROM Table.name
WHERE ID = 1
```

1.6 Personal DBs:

These are typically just local DBs and tables stored on the system. They can be found with the localhost flags. A crucial part of local DBs is that they only support one connection at a time.

1.7 Commercial DBs:

Multiple connections with many things happening at once. Connections are usually facillitated through JAVA or C#.

2 SQL Part 1

2.1 Keys:

2.1.1 Primary

When creating a table in SQL it must have a unique identifier. For example:

EmployeeID	FirstName
1	Lachlan
2	Paul
3	Paul

2.1.2 Composite

Sometimes a comp PK is utilized to define N:M relational tables. As long as the pair of keys is unique, the table will not break. For example:

ProjectID	EmployeeID
1200	1
1300	1
1300	2
1400	2

2.1.3 Foreign

These keys allow relationships between tables to form. The link must be done back to a PK in the other table. When setting a FK during the creation of the table there are two points to understand:

1. Cascade Delete: Once the original PK is deleted all rows with it as a FK
2. Cascade Update: Same as delete, but with updating all FK rows

2.2 SELECT Clause:

Used to select rows of data from a table. There are only two compulsary parts to the statements. That being the column names and table name or names. To remove duplicate rows doing a SELECT DISTINCT statement will remove duplicate rows.

2.3 WHERE Clause:

Filters rows by constraining items that are in the column name. For example:
SELECT EmployeeName
FROM employee
WHERE EmployeeID = 1

Further constraints can be added to the WHERE clause. Such as, IN, NOT, ALL, BETWEEN, IS NULL/IS NOT NULL, and LIKE. The LIKE can be defined with % for infinite number of characters and _ for 1 character. The _ can be done multiple times (e.g. _ _ _, but joined together like ___) for a set defined number of characters.

2.4 Ordering Rows:

Use the ORDER BY after the WHERE. Ordering can either be defined DESC or ASC, with most DBMSs having ASC as the default.

2.5 Having:

This is basically just a WHERE clause for grouped functions. When in a statement with WHERE clauses, the WHERE clause is applied before the HAVING.

2.6 Subqueries:

```
SELECT *
FROM table1
WHERE column IN (subquery)
```